

Joo-Kyung Kim

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PROFESSIONAL EXPERIENCE

Amazon

Senior Applied Scientist

Bellevue, WA

Aug 2017 – Present

- Building core LLM components powering Alexa for hundreds of millions of users — specializing in LLM post-training, multi-turn/proactive interaction, personalization, and evaluation/reward modeling.
- Founded and tech-leading a system and LLM post-training for multi-turn/proactive interaction:
 - * Achieved 3x increase in active users and 2x higher engagement rate.
- Tech-led disambiguation and clarification across multi-turn interactions:
 - * Reduced millions of customer-facing defects per week via disambiguation/clarification.
- Drove hypothesis ranking for large-scale natural language understanding:
 - * Shipped the first production system enabling dynamic routing across thousands of Alexa skills.

Microsoft

Research Intern

Bellevue, WA

May 2016 – Aug 2016

- Deep learning on conversational understanding (Mentors: Gokhan Tur, Asli Celikyilmaz)

NEC Laboratories

Research Associate

Princeton, NJ

May 2015 – Aug 2015

- Constituency parsing with deep and recursive neural networks (Mentor: Bing Bai)

Nuance

Research Intern

Sunnyvale, CA

Jun 2014 – Aug 2014

- Named entity recognition with neural networks for spoken dialog systems (Mentor: Adwait Ratnaparkhi)

Naver

Manager

Seongnam, Korea

Sep 2008 – Jul 2010

- Web search modeling for Naver Japan, and developed integrated Naver search services (C++, PHP, MySQL)

Zio Interactive

Software Engineer (Alternative military service)

Seoul, Korea

Oct 2002 – Mar 2004

- Built 3 game servers and P2P frameworks (C++) and 3 mobile game clients (Brew, Java ME)

IBK System

Software Engineer (Alternative military service)

Seoul, Korea

Feb 2001 – Oct 2002

- Built middleware, TP-monitor, and financial risk management systems (C, C++, Delphi)

EDUCATION

The Ohio State University

Ph.D., Department of Computer Science and Engineering

Columbus, OH

Aug. 2017

- Dissertation: Linguistic Knowledge Transfer for Enriching Vector Representations
- Advisor: Eric Fosler-Lussier

Seoul National University

M.S., School of Computer Science and Engineering

Seoul, Korea

Aug. 2008

- Thesis: Evolutionary optimization of a collection of variable-length subpatterns for pattern classification
- Advisor: Byoung-Tak Zhang

Sogang University

B.E., Department of Computer Science

Seoul, Korea

Aug. 2005

- Thesis: Real-time news categorization system
- Advisor: Jihoon Yang

SELECTED PUBLICATIONS

- Jinyoung Park, Sanghyeok Lee, Omar Zia Khan, Hyunwoo J. Kim, and **Joo-Kyung Kim**. *HyperGraphPro: Progress-Aware Reinforcement Learning for Structure-Guided Hypergraph RAG*. 2026. arXiv: 2601.17755.
- Soyeong Jeong, Taehee Jung, Sung Ju Hwang, **Joo-Kyung Kim**, and Dongyeop Kang. “When Thoughts Meet Facts: Reusable Reasoning for Long-Context LMs”. In: *ACL Findings*. 2026.
- Qiushi Zhan, Angeline Budiman-Chan, Abdelrahman Zayed, Xingzhi Guo, Daniel Kang, and **Joo-Kyung Kim**. “SafeSearch: Do Not Trade Safety for Utility in LLM Search Agents”. In: *EACL Findings*. 2026.
- Zae Myung Kim, Anand Ramachandran, Farideh Tavazoei, **Joo-Kyung Kim**, Oleg Rokhlenko, and Dongyeop Kang. “Align to Structure: Aligning Large Language Models with Structural Information”. In: *AAAI*. 2026.
- Chanwoo Park, Seungju Han, Xingzhi Guo, Asuman Ozdaglar, Kaiqing Zhang, and **Joo-Kyung Kim**. “MAPoRL: Multi-Agent Post-Co-Training for Collaborative Large Language Models with Reinforcement Learning”. In: *ACL*. 2025.
- Shirley Anugrah Hayati, Taehee Jung, Tristan Bodding-Long, Sudipta Kar, Abhinav Sethy, **Joo-Kyung Kim**, and Dongyeop Kang. “Chain-of-Instructions: Compositional Instruction Tuning on Large Language Models”. In: *AAAI*. 2025.
- Jinyoung Park, Minseok Joo, **Joo-Kyung Kim**, and Hyunwoo J. Kim. “Generative Subgraph Retrieval for Knowledge Graph-Grounded Dialog Generation”. In: *EMNLP*. 2024.
- Jihyung Kil, Farideh Tavazoei, Dongyeop Kang, and **Joo-Kyung Kim**. “II-MMR: Identifying and Improving Multi-modal Multi-hop Reasoning in Visual Question Answering”. In: *ACL Findings*. 2024.
- Jinyoung Park, Ameen Patel, Omar Zia Khan, Hyunwoo J. Kim, and **Joo-Kyung Kim**. *Graph Elicitation for Guiding Multi-Step Reasoning in Large Language Models*. 2023. arXiv: 2311.09762.
- Taehee Jung, **Joo-Kyung Kim**, Sungjin Lee, and Dongyeop Kang. “Cluster-Guided Label Generation in Extreme Multi-Label Classification”. In: *EACL*. 2023.
- Joo-Kyung Kim**, Guoyin Wang, Sungjin Lee, and Young-Bum Kim. “Deciding Whether to Ask Clarifying Questions in Large-Scale Spoken Language Understanding”. In: *ASRU*. 2021.
- Joo-Kyung Kim** and Young-Bum Kim. “Pseudo Labeling and Negative Feedback Learning for Large-scale Multi-label Domain Classification”. In: *ICASSP*. 2020.
- Joo-Kyung Kim** and Young-Bum Kim. “Supervised Domain Enablement Attention for Personalized Domain Classification”. In: *EMNLP*. 2018.
- Joo-Kyung Kim** and Young-Bum Kim. “Joint Learning of Domain Classification and Out-of-Domain Detection with Dynamic Class Weighting for Satisficing False Acceptance Rates”. In: *Interspeech*. 2018.
- Young-Bum Kim, Dongchan Kim, **Joo-Kyung Kim**, and Ruhi Sarikaya. “A Scalable Neural Shortlisting-Reranking Approach for Large-Scale Domain Classification in Natural Language Understanding”. In: *NAACL*. 2018.
- Joo-Kyung Kim**, Young-Bum Kim, Ruhi Sarikaya, and Eric Fosler-Lussier. “Cross-Lingual Transfer Learning for POS Tagging without Cross-Lingual Resources”. In: *EMNLP*. 2017.
- Joo-Kyung Kim**, Gokhan Tur, Asli Celikyilmaz, Bin Cao, and Ye-Yi Wang. “Intent Detection using Semantically Enriched Word Embeddings”. In: *SLT*. 2016.
- Joo-Kyung Kim**, Marie-Catherine de Marneffe, and Eric Fosler-Lussier. “Adjusting Word Embeddings with Semantic Intensity Orders”. In: *ACL Workshop on Representation Learning for NLP (RepL4NLP)*. 2016.
- Joo-Kyung Kim**, Marie-Catherine de Marneffe, and Eric Fosler-Lussier. “Neural word embeddings with multiplicative feature interactions for tensor-based compositions”. In: *NAACL Workshop on Vector Space Modeling for NP (VSM)*. 2015.
- Joo-Kyung Kim** and Marie-Catherine de Marneffe. “Deriving adjectival scales from continuous space word representations”. In: *EMNLP*. 2013.
- Joo-Kyung Kim** and Byoung-Tak Zhang. “Evolving hypernetworks for pattern classification”. In: *IEEE Congress on Evolutionary Computation (CEC)*. 2007.
- Joo-Kyung Kim**, Byung Soo Kim, Oh Kyuk Kwon, Seung Kon Hwang, Jung-Woo Ha, Chan-Hoon Park, Duck Jin Chung, Chong Ho Lee, Jaehyun Park, and Byoung-Tak Zhang. “A DNA computing-inspired silicon chip for pattern recognition”. In: *International Meeting on DNA Computing (DNA)*. 2007.
- Byoung-Tak Zhang and **Joo-Kyung Kim**. “DNA hypernetworks for information storage and retrieval”. In: *International Meeting on DNA Computing (DNA)*. 2006.

ACADEMIC SERVICES

Senior Area Chair

- ARR 2025-2026, EMNLP 2025-2026, ACL 2026

Area Chair

- ARR Dec 2023 - Feb 2025

Reviewer

- ACL 2019-2023, EMNLP 2019-2023, NAACL 2019-2022, EACL 2021-2023, ACL 2023, AAAI 2020-2022, IEEE TASLP 2020-2021

SKILLS

Programming Languages

- Python, JAVA, C/C++, Lua, Objective C, Delphi, PHP

Machine Learning Frameworks

- PyTorch, Torch, DyNet, Matlab